Project 2:

E-Commerce Customer Segmentation and Product Recommendation

Problem Definition

E-commerce platforms generate large amounts of customer and transaction data. However, identifying meaningful customer groups and recommending relevant products is challenging without labeled data.

The problem is to:

- Segment customers based on behavior
- Recommend products based on purchase patterns

using unsupervised learning techniques.

Project Description

This project implements a data-driven e-commerce analysis system that combines:

1. Customer Segmentation using clustering algorithms
2. Product Recommendation using association rule mining

The system analyzes customer purchase behavior and:

- Groups similar customers into clusters
- Generates product recommendations based on frequently bought items

After comparing multiple clustering algorithms, **DBSCAN** provided the best segmentation performance and was selected as the final model.

Tools & Libraries Used

Programming Language:

- Python

Libraries:

- Streamlit → Web dashboard
- Pandas → Data processing
- NumPy → Numerical operations
- Scikit-learn → Clustering algorithms (DBSCAN)
- MLxtend → Association rule mining

Code Implementation:

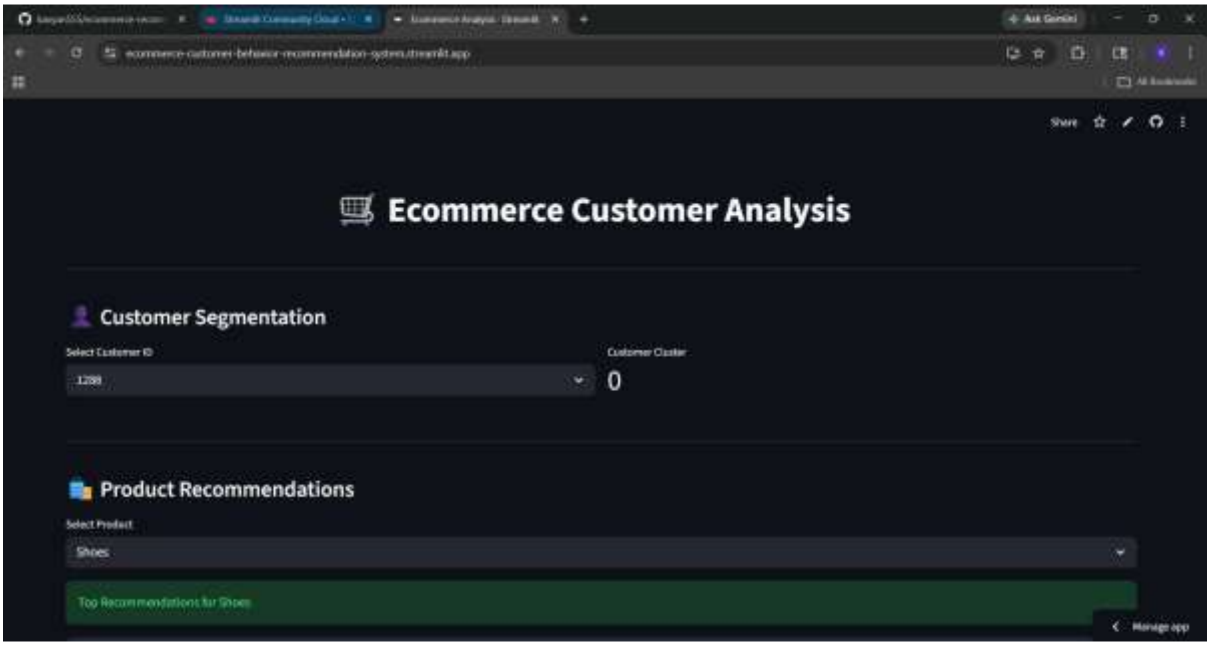
```
import streamlit as st
import pandas as pd

# PAGE CONFIG
st.set_page_config(page_title="Ecommerce Analysis", layout="wide")

# LOAD DATA
df = pd.read_csv("final_data.csv")
rules = pd.read_csv("rules.csv")

# CLEAN FUNCTION
def clean_items(x):
    try:
        items = list(eval(x))
        return ", ".join(items)
    except:
        return x

rules['antecedents'] = rules['antecedents'].astype(str)
rules['consequents'] = rules['consequents'].astype(str)

rules['antecedents_clean'] = rules['antecedents'].apply(clean_items)
rules['consequents_clean'] = rules['consequents'].apply(clean_items)

# TITLE
st.markdown("<h1 style='text-align:center;'>🛒 Ecommerce Customer Analysis</h1>",
unsafe_allow_html=True)
st.write("---")
```

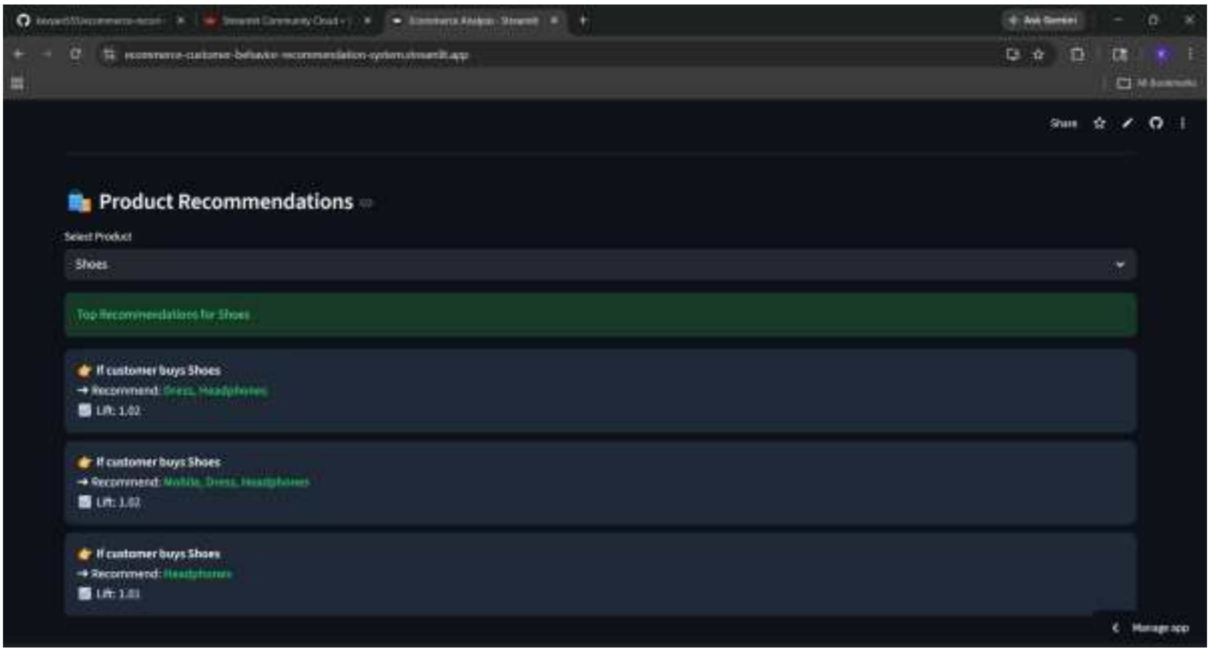
```
# CUSTOMER SEGMENTATION
st.subheader("👤 Customer Segmentation")

col1, col2 = st.columns(2)

with col1:
    customer_id = st.selectbox("Select Customer ID", df['CustomerID'].unique())

with col2:
    if 'Cluster' in df.columns:
        cluster = df[df['CustomerID'] == customer_id]['Cluster'].values[0]
        st.metric(label="Customer Cluster", value=cluster)
    else:
        st.warning("Cluster data not available")

st.write("---")

# RECOMMENDATION SYSTEM
st.subheader("🛍️ Product Recommendations")

product = st.selectbox("Select Product", df['Product'].unique())

# FILTER RULES

filtered = rules[
    rules['antecedents_clean'].str.contains(product, case=False, na=False)
]

# Remove duplicate recommendations
filtered = filtered.drop_duplicates(subset=['consequents_clean'])
```

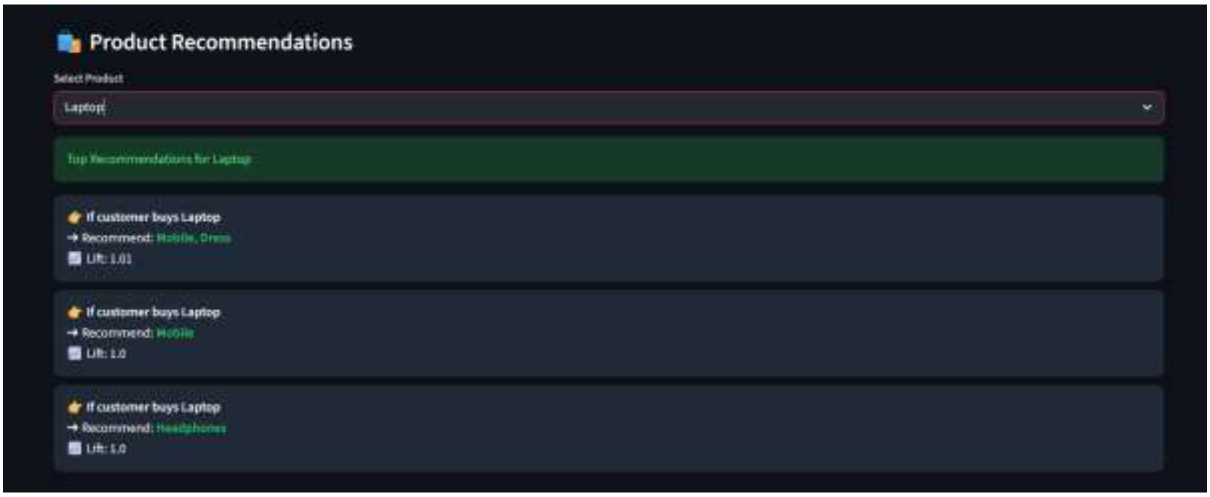
```
# Sort by lift (best first)
filtered = filtered.sort_values(by='lift', ascending=False)

# Take top 3
top_rules = filtered.head(3)

# DISPLAY RESULTS
if not top_rules.empty:
    st.success(f"Top Recommendations for {product}")

    for _, row in top_rules.iterrows():
        st.markdown(f"""
        <div style="
            background-color:#1f2937;
            padding:15px;
            border-radius:10px;
            margin-bottom:10px;
        ">
        👉 <b>If customer buys {product}</b><br>
        ➜ Recommend: <span style="color:#22c55e;"><b>{row['consequents_clean']}</b></span><br>
        📈 Lift: {round(row['lift'],2)}
        </div>
        """, unsafe_allow_html=True)

else:
    st.warning("No recommendations found")

st.write("---")

# OPTIONAL DATA VIEW
```

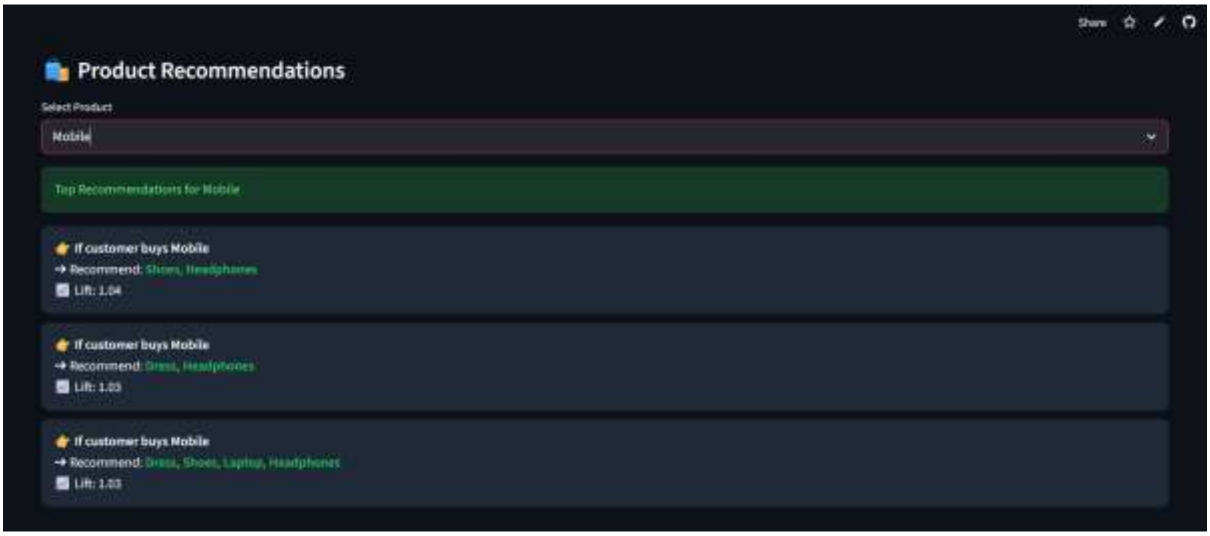
```
with st.expander("📊 View Sample Data"):
    st.dataframe(df.head())
```

Output:






Product Recommendations
Select Product
Laptop
Top Recommendations for Laptop
If customer buys Laptop
→ Recommend: Mobile, Dress
Lift: 1.01
If customer buys Laptop
→ Recommend: Mobile
Lift: 1.0
If customer buys Laptop
→ Recommend: Headphones
Lift: 1.0


Share
Product Recommendations
Select Product
Mobile
Top Recommendations for Mobile
If customer buys Mobile
→ Recommend: Shoes, Headphones
Lift: 1.04
If customer buys Mobile
→ Recommend: Dress, Headphones
Lift: 1.03
If customer buys Mobile
→ Recommend: Dress, Shoes, Laptop, Headphones
Lift: 1.03


Share
Product Recommendations
Select Product
Mobile
Dress
Headphones
Laptop
Mobile
Shoes
If customer buys Mobile
→ Recommend: Dress, Headphones
Lift: 1.03
If customer buys Mobile
→ Recommend: Dress, Shoes, Laptop, Headphones
Lift: 1.03